Summary

- DNA Replication : Each strand of the DNA double helix serves as a template for the synthesis of a complementary daughter strand by the semiconservative mode of replication which was confirmed by the Meselson Stahl experiment.
- The two strands of the double helix unwind with the help of helicase. DNA, being a large molecule, the separation of the two single strands creates a 'Y' shaped structure called a 'replication fork'. DNA polymerases synthesize the new DNA strands only in the 5' to 3' direction.Therefore, one of the newly synthesized strands grows in the 5' to 3' direction towards the replication fork (leading strand) and and one in the 5' to 3' direction away from the replication fork (lagging strand).
- A short stretch of RNA acts as a primer for DNA polymerase and it is synthesized by RNA primase.The leading strand needs only one primer, but the lagging strand needs many and short fragments of DNA called Okazaki fragments are synthesized, which are eventually ligated.
- In *E.coli*, DNA polymerase III elongates a DNA chain and proofreads mismatched bases with its 3' to 5' exonuclease activity. DNA polymerase I removes RNA primers using its 5' to 3' exonuclease activity and fills the gaps.
- Transcription: The process of RNA synthesis directed by a DNA template is called transcription and it is performed by RNA polymerase. Transcription occurs in 3 steps: initiation, elongation and termination.
- RNA polymerase binds to a sequence of DNA called the promoter, found near the start of a gene. Once bound, RNA polymerase separates the DNA strands, exposing the single-stranded DNA template. Complementary bases are added (instead of T, U is added) one at a time by RNA polymerase.The newly synthesized RNA strand grows from 5' to 3'. Termination occurs with the help of a rho factor or with specific terminator sequences.
- In eukaryotes post transcriptional modifications occur by which, a methyl guanosine cap and a poly A tail are added to 5' and 3' ends, respectively, to mRNA. Introns are spliced out of mRNA, tRNA and rRNA primary transcripts.
- Translation: The three major components in translation are mRNA, ribosomes and tRNA. The mRNA code is read in the 5' to 3' direction, and the protein is synthesized from the amino terminus to the carboxyl terminus.Three steps occur during translation: Initiation, elongation and termination.
- There is at least one specific tRNA and one aminoacyl tRNA synthetase for each amino acid. An anticodon in tRNA recognizes the codon present in mRNA.
- Initiation of translation: mRNA associates with the small ribosomal subunit and this process requires a sequence called Shine-Dalgarno sequence on mRNA.

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